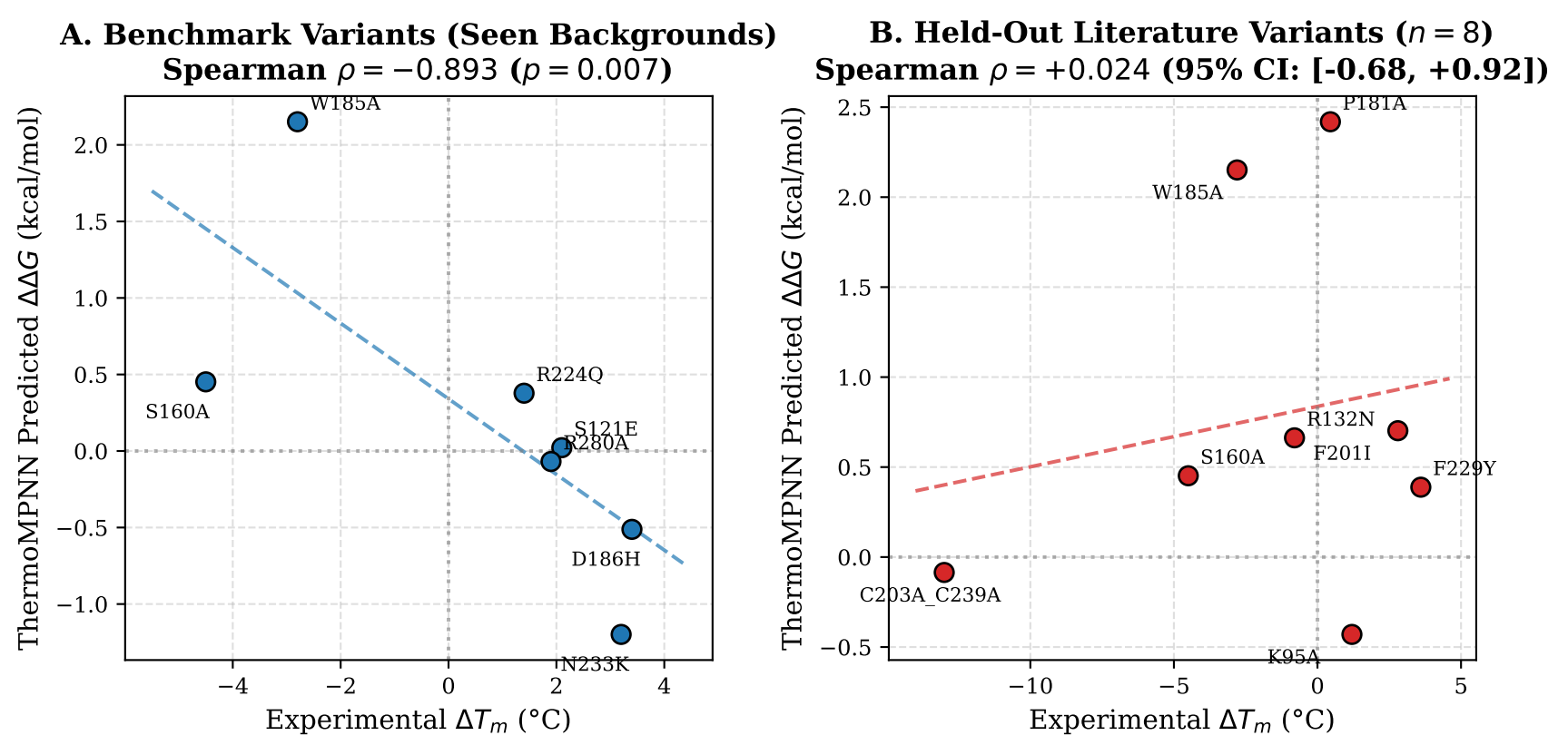
A. Benchmark Variants (Seen Backgrounds)
Spearman ρ = −0.893 (p = 0.007)
ThermoMPNN Predicted ΔΔG (kcal/mol)
Experimental ΔTm (°C)
W185A
S160A
R224Q
S121E
R280A
D186H
N233K
B. Held-Out Literature Variants (n = 8)
Spearman ρ = +0.024 (95% CI: [-0.68, +0.92])
ThermoMPNN Predicted ΔΔG (kcal/mol)
Experimental ΔTm (°C)
P181A
W185A
R132N
F201I
S160A
F229Y
C203A_C239A
K95A